

# Attack–Fault Confounding in a Linear Model of the Electronic Trading Gateway Is Generically Fragile under State-Level Injection

Chiheb Mejrissi<sup>a,\*</sup>

*<sup>a</sup>Department of Computer Science, Example University, 1 Example Street, Example City, Country*

---

## Abstract

Trading-gateway operators must act on suspicious order flow without knowing whether a compromise or a software fault caused it. In a linear-Gaussian model of gateway-observable order flow, the causes are indistinguishable if and only if their projected injections coincide, and ambiguity then has a strictly positive cost; both specialise classical results. Once the incident feeds into the dynamics, confounding survives only if the injection difference is Kalman-unobservable: for almost every dynamics matrix, not necessarily for structured market dynamics, only identical state injections, or identically distributed random ones, confound. For a pair confounded only at the output, the cost of ambiguity stays maximal at every horizon under output-level injection but falls below it within  $n$  post-onset observations under state-level injection if  $(A, C)$  is observable. In four ABIDES designs where a one-parameter search tunes the fault toward a fixed compromise on one summary statistic, with no pre-specified tolerance, the summary-matched regime’s accuracy interval on retrospective complete-episode features excluded one half in every design, also after multiplicity adjustment, and exceeded a small negative control’s bias bound in three; the fourth is indeterminate. At one illustrative cost triple, with a fault loss modelled from measured submitted volume, a classifier-based policy beats the best constant policy with a supported interval in 40 of 50 cells from 10 profile–regime units (39 after max-T adjustment) under the primary convention and otherwise

---

\*Corresponding author.

*Email address:* `chiheb.mejrissi@example.org` (Chiheb Mejrissi)

ties or is inconclusive; against the best single-statistic threshold rule, 1 cell is a supported deterioration under the rebaselined convention, inconclusive after adjustment.

*Keywords:* cyber-physical security, electronic trading, attack-fault identifiability, cost of ambiguity, decision under uncertainty, incident response, agent-based market simulation

---

## 1. Introduction

A trading participant can generate harmful or suspicious orders because of a compromised credential, a software defect, its own order-generation policy, ordinary market conditions, or a combination of these. An operator — the incident-response controller of a broker or trading firm — must often act before the cause is known: throttle, cancel orders, revoke access, or isolate a process. Each action has a cost when applied to a case that did not need it, and each carries risk when withheld from a case that did. This paper answers two questions that come before when to act (timing is out of scope; Section 3.5).

First: **can the causes be told apart** from the evidence an operator is ordinarily permitted to observe (order acknowledgments, market data, session and process telemetry)? Second, if not: **how costly is that ambiguity**, measured by how much the best achievable expected cost exceeds that of an oracle that knows the cause? A threshold-based kill switch answers neither question: it trips on an order rate, a cancel ratio or a loss limit and acts, with no attempt to establish cause or to price the error.

This paper answers the first question with a formal identifiability result (Sections 3–4), scoped to a linear-Gaussian order-flow model, the setting of the classical fault-diagnosis and observability results it specialises (Section 2). The cost of ambiguity is defined with the loss table (Definition D7, Section 3.4); Sections 5–6 check the theory and estimate that cost in a linear pilot. Section 7 studies retrospectively, in the ABIDES agent-based market simulator, what attempting to match two incident mechanisms on a chosen summary statistic does to classification and to decision cost (Section 7.1).

The paper’s thesis is that, in the linear model, exact attack/fault confounding is a knife-edge. It needs equal observation laws, here equal projected injections (Theorem 1): easy to meet under output-level injection with few observed channels (Corollary 1), generically broken under state-

level injection, when the incident propagates through the book’s dynamics (Section 4.1). The simulator study does not test this fragility. It tests the premise of the model-free Proposition 2, under which approximate confounding keeps the cost of ambiguity near its maximum: two observation laws close in total variation. Tuning a fault toward a compromise on one summary statistic, which does not equalize laws even in a linear model (Section 7.1), left them far apart in three of four designs, by a classifier’s lower accuracy limit (Corollary 5), and the fourth indeterminate. The bridge runs one way: a cell at chance could not have confirmed the premise.

### **Contributions.**

1. A comparison of one compromise/fault pair under output-level and state-level injection. Its statements specialise classical observability [1, 2], as Theorem 1 specialises fault-mode distinguishability: output-level confounding survives state-level injection only on the Kalman unobservable subspace (Theorems 2–3, the attack-free case of [3]; Corollaries 3–4), also for random injections of arbitrary law (Theorem 4). The contribution is the comparison, put in cost form by Proposition 4: for a pair Theorem 1 confounds whose state injections differ, the cost of ambiguity stays at its maximum at every horizon under output-level injection and falls below it from a horizon  $h^* \leq n$  under state-level injection with observable dynamics. In a simulation, an exactly confounded pair becomes classifiable under one stable, coupled, observable  $A$  while it stays at chance under output-level injection.
2. An ABIDES study of what a one-parameter calibration toward one summary statistic leaves distinguishable (Section 7). In four injection designs, a search that tunes only the fault’s magnitude (left unscaled in D1 and D2) toward a fixed compromise, without reaching a pre-specified tolerance, left the summary-matched regime above chance in every design and beyond the pipeline bias a negative control leaves open in three. The cells do not separate leakage through channels the search left unmatched from a residual difference on the target (Section 7.2).

## **2. Related work**

**Decision-theoretic cyber response.** POMDP formulations of cyber defense choose actions under noisy alerts as an attacker progresses [4]; related work uses Stackelberg games over attack-response trees [5], balances service

continuity against intrusion risk [6] (with learning-based extensions [7, 8]), or supports decisions with models [9]. None of it prices market impact; our loss table uses a market-impact proxy, not a model of liquidity replacement.

**Cyber-physical fault-versus-attack differentiation.** The closest analog separates physical faults from cyberattacks; its classical core is fault diagnosis: two failure modes are separable when their output signatures differ [10], and in stochastic linear models fault-mode distinguishability is a divergence between output laws, zero exactly when they coincide [11]; Theorem 1 is this criterion for one compromise and one fault instance. In secure control, a divergence-based stealth bound limits every detector [12]. Pasqualetti, Dörfler and Bullo [3] characterize which attacks are undetectable or unidentifiable by an output-based monitor (those that excite only the zero dynamics); Theorem 2 uses the attack-free special case of this object, the Kalman unobservable subspace. Roy and Dey [13] derive, with an unknown-input observer, a closed-form fault/attack distinguishability criterion whose fault-mimicking condition is the analog of our Corollary 1. Observer-based fault and attack isolation schemes [14, 15] address the design side, and two recent designs use the closed loop itself: Zhou et al. [16] pair a plant-side fault detector with an observer-based attack detector, because multiplicative faults and additive attacks act differently on the closed-loop dynamics, and Xue et al. [17] add a plant-side check of the controller’s input–output consistency, which kernel (zero-dynamics, covert, replay) attacks excite and faults do not. These synthesise detectors for given fault and attack classes, and none prices the response or treats trading. Theorems 2–4 and Corollaries 3–4 synthesise nothing, and only Corollary 4 takes the dynamics matrix as generic, so that, unless the two state injections are identical, output-level confounding survives state-level injection only on a Lebesgue-null set of dynamics matrices. In active fault diagnosis, inputs are designed over a finite horizon to minimise an upper bound on the probability of choosing the wrong linear model [18, 19], and Bayesian diagnosability is a windowed separation of predicted outputs whose growth drives identification [20]. Proposition 4 designs no input and uses no bound: it evaluates the classical two-point Bayes risk (Proposition 3) on a passive window, to compare the two injection levels of one pair. Data-driven work separates simulated faults from attacks with a classifier [21] or by fusing network and process data [22]. Closest to our framing, the triage framework of Ma et al. for energy systems attributes anomalies to physical faults or to measurement-integrity loss, prices misattribution against blanket responses, and observes that exactly matched

fault/attack pairs are discriminated at chance [23]. It derives no necessary-and-sufficient indistinguishability condition, cost floor or state-level fragility, and does not address trading. Its matched pairs are the empirical counterpart of our Theorem 1 and Proposition 1; Section 7 matches only summary statistics.

Linear stealth need not survive nonlinearity: stealthy injections must be redesigned for nonlinear grid models [24], and on Lie groups an attack keeps impact and stealth across operating conditions only if it commutes with the flow [25]. No analogous condition is known for our order-flow model; we claim neither that it holds nor that it fails.

**Market manipulation detection and trading kill switches.** A large literature detects manipulative patterns (spoofing, wash trading) from trade and order-book data [26]; it neither separates a compromised gateway from a faulty one nor prices the response. Compromised brokerage accounts can automate manipulative order generation [27], which supports modelling a compromise as a structured perturbation of order flow. A threshold trigger on an order-flow statistic is an observation-dependent policy, not one of this paper’s constant reference policies (Section 9).

**Costly information acquisition.** Work on asset markets [28] concerns traders’ incentives to learn about asset value. Closer in method, optimal stopping with costly sequential information studies when to stop paying for signals before a reversible or irreversible decision [29]. This paper buys no information: it bounds what passive observation of order flow can achieve, and leaves an action on a second channel to separate work (Section 7.5).

**Scope of the contribution.** Theorem 1 specialises classical results on the distinguishability of fault modes [10, 11], Theorems 2–4 and Corollaries 3–4 Kalman observability [1, 2], and Propositions 1–4 the Bayes risk of a two-point test [30, 31], to the gateway’s loss table; no statement is new as mathematics. Section 1 lists what we add; it poses the compromise-versus-fault question at a trading gateway, with a market-impact loss table rather than a detection-accuracy metric. Methodologically, the classification test is a classifier two-sample test [32], and the cost comparison is an expected-cost comparison with constant policies over a prior grid.

### 3. Model

Throughout,  $K$  (“compromise”) and  $F$  (“fault”) name fully specified perturbations of the order-flow process — a directional bias, an elevated or-

der rate or an altered cancellation pattern, with a chosen severity and onset — not claims about real attackers or real defects. The results concern *behavioural-mechanism identification*, whether two specified generative processes can be told apart from gateway-observable order flow, not inference about intent or authorization, which would need other evidence (credential logs, code-change records).

### 3.1. Order-flow dynamics

Let  $x_t \in \mathbb{R}^n$  be a latent representation of normal order-flow state (for example, components loading onto price pressure, volume intensity and cancellation propensity), evolving under known linear-Gaussian dynamics with transition matrix  $A$  and process-noise covariance  $Q$ . The complete model is Definition D2 below.

### 3.2. Incident injection

At an onset time  $\tau$ , possibly unknown to the operator, an incident of mechanism  $M \in \{K, F\}$  injects an additive perturbation into the pre-observation order-flow signal, with direction  $b_M$  and severity  $\theta_M$ .

**Definition D1 (Mechanism instance).** A mechanism instance is a triple  $(M, \theta_M, b_M)$  with cause  $M \in \{K, F\}$  ( $K$ : compromise,  $F$ : fault), a deterministic severity  $\theta_M \in \mathbb{R}$  and a deterministic injection direction  $b_M \in \mathbb{R}^n$ . Statements compare one fixed instance of  $K$  with one fixed instance of  $F$  (pairwise-instance reading); no distribution over severities, directions or onsets within a cause is modelled.

### 3.3. Gateway-observable output

The operator observes a noisy projection of order flow, typically lower-dimensional ( $m < n$ ), through a fixed, known matrix  $C \in \mathbb{R}^{m \times n}$ , representing what a gateway or market-data feed surfaces (spread, depth, cancellation rate). In this model the incident shifts the signal at the point of observation only; Section 4.1 lets it propagate through the dynamics.

**Definition D2 (Output-level injection model).** Fix  $n, m$ , a matrix  $A \in \mathbb{R}^{n \times n}$ , an observation matrix  $C \in \mathbb{R}^{m \times n}$ , covariance matrices  $Q \in \mathbb{R}^{n \times n}$  and  $\Sigma \in \mathbb{R}^{m \times m}$ , and an onset time  $\tau$ . Under a mechanism instance  $(M, \theta_M, b_M)$  (Definition D1), the latent state, the perturbed state and the observation evolve for  $t \geq 0$  as

$$\begin{aligned} x_{t+1} &= Ax_t + w_t, & \tilde{x}_t &= x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, \\ y_t &= C\tilde{x}_t + \varepsilon_t, \end{aligned}$$

with  $w_t \sim \mathcal{N}(0, Q)$  and  $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ . The injection enters the observation only; the state recursion does not see it.

Everything except the injection term is common to the two causes. Section 4 rests on the following standing assumptions, which the implementation satisfies; the finite-horizon experiments of Sections 5–6 always include time steps after the onset, so (A5) holds.

**Standing assumptions.**

**(A1) Gaussian, independent noise.**  $w_t \sim \mathcal{N}(0, Q)$  and  $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ , with  $Q$  and  $\Sigma$  symmetric positive semidefinite; the family  $\{w_t\}_{t \geq 0} \cup \{\varepsilon_t\}_{t \geq 0}$  is mutually independent (across time and across the two sequences), and its law does not depend on the instance.

**(A2) Deterministic initial state.**  $x_0 = 0$ . (The proofs use only that  $x_0$  is Gaussian, possibly degenerate, independent of the noise, with a law that does not depend on the instance;  $x_0 = 0$  is the implemented special case.)

**(A3) Known, common model.**  $A, Q, C, \Sigma$  are known and identical under both causes; only the injection term  $\theta_M b_M$  depends on the instance.

**(A4) Common onset.** The onset  $\tau$  is deterministic, common to both causes, not necessarily known to the operator. (The proofs use only that  $\tau$  is deterministic and common; no statement requires a policy to know  $\tau$ .)

**(A5) Window reaches the onset.** The observation window  $W$  contains at least one  $t \geq \tau$ . Theorem 2 uses the stronger form  $W \supseteq \{t : t \geq \tau\}$  and Theorem 3 the stronger form  $W \supseteq \{\tau, \dots, \tau + H - 1\}$ .

**(A6) Positive-definite observation noise.**  $\Sigma \succ 0$ . Used only to apply Proposition 3 to the model on a finite window: it makes the covariance of the stacked observations positive definite ( $V \succ 0$ ). No other statement uses a positivity condition on  $\Sigma$  or  $Q$ ;  $Q$  is only required to be positive semidefinite.

**(A7) Ordered loss table.**  $H_K > C_K$  and  $C_F > H_F$  (Definition D6): leaving a compromise alone costs strictly more than containing it, and containing a fault costs strictly more than leaving it alone. Propositions 1 and 2 carry it as their hypotheses  $\text{HARM} > C_{OP}$  and  $\text{DISRUPT} > 0$ ; Propositions 3 and 4 as  $a, b > 0$ . No identifiability statement (Theorems 1–4, Corollaries 1–4) uses it. Every use of the threshold  $q^* = (C_F - H_F) / [(H_K - C_K) + (C_F - H_F)]$ , of  $R_{\text{oracle}} = p C_K + (1 - p) H_F$  and

of the best constant policy’s excess  $\min\{p(H_K - C_K), (1-p)(C_F - H_F)\}$  over the oracle requires it, so an empirical loss table to which these are applied must satisfy it in every cell.

**Definition D3 (Observation window; observational indistinguishability).** An observation window is a set  $W \subseteq \{0, 1, 2, \dots\}$  of time indices, finite or not; the observation trajectory on  $W$  is  $Y_W = (y_t)_{t \in W}$ . Write  $P_M^W$  for the law of  $Y_W$  under instance  $(M, \theta_M, b_M)$  (for infinite  $W$ , the law on the product  $\sigma$ -algebra, determined by its finite-dimensional marginals). Instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  are observationally indistinguishable on  $W$  if  $P_K^W = P_F^W$ . Theorem 1 uses  $W = \{t : t \geq 0\}$ ; “indistinguishable at every  $t \geq \tau$ ” (Theorem 2) means  $P_K^W = P_F^W$  for  $W = \{t : t \geq \tau\}$ ; Theorem 3 (the finite-horizon version of Theorem 2) uses  $W = \{\tau, \dots, \tau + H - 1\}$ .

**Definition D4 (Distinguishable in the population limit).** Two instances are distinguishable in the population limit on  $W$  if they are not observationally indistinguishable on  $W$ , i.e.  $P_K^W \neq P_F^W$  (equivalently  $\text{TV}(P_K^W, P_F^W) > 0$ , Definition D8). This is a statement about the two laws only; it asserts nothing about the accuracy attainable from finitely many observations.

**Definition D5 (Gaussian discriminability).** If  $Y \mid M \sim \mathcal{N}(\mu_M, V)$ ,  $M \in \{K, F\}$ , with a common covariance  $V \succ 0$ , let  $\delta = \mu_K - \mu_F$  and  $d = \sqrt{\delta^\top V^{-1} \delta} \geq 0$  (the whitened mean separation).

#### 3.4. The loss table, and what it assumes

The cost analysis uses the two-action, two-cause loss table

| Action      | Compromise $K$ | Fault $F$ |
|-------------|----------------|-----------|
| Contain     | $C_{OP}$       | DISRUPT   |
| Leave alone | HARM           | 0         |

a special case of the general table below; Proposition 1 keeps these symbols.

**Definition D6 (Loss table).** The general two-action, two-cause loss table  $L(a, M)$ ,  $a \in \{\text{contain, leave alone}\}$ ,  $M \in \{K, F\}$ , is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave alone}, K) &= H_K, & L(\text{leave alone}, F) &= H_F, \end{aligned}$$

with  $C_K, C_F, H_K, H_F \in \mathbb{R}$ . This paper’s table (the two-action table printed above this definition) is the special case

$$C_{OP} = C_K, \quad \text{DISRUPT} = C_F, \quad \text{HARM} = H_K, \quad H_F = 0.$$

Propositions 1 and 2 are stated in this paper’s symbols  $C_{OP}$ , DISRUPT, HARM.

**Definition D7 (Prior, policies, Bayes risk, oracle risk, cost of ambiguity).** The cause is drawn as  $M = K$  with prior probability  $p = P(K)$  and  $M = F$  with probability  $1 - p$ ; given  $M$ , the observation trajectory  $Y_W$  has law  $P_M^W$ . A policy is a measurable map  $\pi$  from observation trajectories on  $W$  to  $[0, 1]$ , read as the probability of the action *contain* (randomized policies allowed; a deterministic policy takes values in  $\{0, 1\}$ ). Its expected cost is

$$R(\pi) = p \mathbb{E}_K [\pi(Y_W) C_K + (1 - \pi(Y_W)) H_K] \\ + (1 - p) \mathbb{E}_F [\pi(Y_W) C_F + (1 - \pi(Y_W)) H_F].$$

The best achievable expected cost is  $R^* = \inf_{\pi} R(\pi)$ ; the oracle, which observes  $M$ , has expected cost  $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$ ; the cost of ambiguity is  $\text{CoA}^* = R^* - R_{\text{oracle}}$ . Where the observation is the raw trajectory  $Y$  the subscript  $Y$  is written ( $R_Y^*$ ,  $\text{CoA}_Y^*$ ).

**Definition D8 (Observation densities; total variation).** Let  $\mu$  be a  $\sigma$ -finite measure dominating  $P_K$  and  $P_F$  (e.g. the finite measure  $P_K + P_F$ ), and  $f_K, f_F$  their densities with respect to  $\mu$ ;  $\int \cdot dy$  denotes integration against  $\mu$ . The total variation distance is

$$\text{TV}(P, P') = \sup_B |P(B) - P'(B)| = \frac{1}{2} \int |f - f'| dy,$$

so that  $1 - \text{TV}(P_K, P_F) = \int \min\{f_K, f_F\} dy$ .

The table encodes a scenario in which leaving a fault alone is free, for example a logging error that inflates apparent order counts. The strict value of distinguishing the two causes is zero whenever one action is optimal for both; Proposition 1’s cost floor is a property of loss structures in which the causes call for different actions. Because the simulated fault mechanisms submit real orders, Section 7.3 replaces the zero entry by a positive fault loss  $H_F$ .

### 3.5. Observation timing and scope

Theorem 1 and Proposition 1 concern the whole observation trajectory, and both the linear pilot and the ABIDES study use whole-episode summary features (mean, standard deviation and net drift of each observed channel), not a prefix available at a pre-committed decision deadline. The empirical study is therefore retrospective diagnosis: a real-time claim would need an

explicit observation cutoff and prefix features, and we make none. The identifiability results, which concern the laws of whole trajectories, are unaffected, but the costs of Sections 6–7 do not measure the value of information at an operational decision point.

#### 4. Theoretical results

Section 2 gives each statement’s classical source; we state them because the rest of the paper uses their exact form, and Table 1 lists what each uses. Appendix A proves Theorem 4, whose point-mass case gives Theorems 1–3 and Corollary 3, Corollary 4 and Proposition 4, citing only the main text; Supplement S1, submitted with the manuscript, proves every numbered statement.

Table 1: What each statement uses: definitions, standing assumptions and earlier results.

| Statement     | Definitions                                             | Assumptions; results used                                           |
|---------------|---------------------------------------------------------|---------------------------------------------------------------------|
| Theorem 1     | D1–D3 ( $W = \{t : t \geq 0\}$ )                        | (A1)–(A5)                                                           |
| Corollary 1   | D1                                                      | condition (*) of Theorem 1; linear algebra only                     |
| Corollary 2   | D1–D4                                                   | (A1)–(A5), through Theorem 1                                        |
| Proposition 1 | D1–D3, D6 (this paper’s symbols), D7                    | (A1)–(A5), through Theorem 1                                        |
| Proposition 2 | D6 (this paper’s symbols), D7, D8                       | none of (A1)–(A6): any two laws                                     |
| Proposition 3 | D5, D6 (general symbols), D7, D8                        | on the model’s finite windows, (A1)–(A6)                            |
| Theorem 2     | D1, D3 ( $W = \{t : t \geq \tau\}$ ), D9, D10           | (A1)–(A4), (A5) in its all-time form                                |
| Theorem 3     | D1, D3 ( $W = \{\tau, \dots, \tau + H - 1\}$ ), D9, D10 | (A1)–(A4), (A5) in its finite-horizon form                          |
| Corollary 3   | as Theorem 2                                            | as Theorem 2                                                        |
| Corollary 4   | as Theorem 2, and D10, D11                              | as Theorem 2; Corollary 3                                           |
| Theorem 4     | D2, D3, D9 ( $s_M$ for $\theta_M b_M$ ), D10            | (A1)–(A5); the three hypotheses on $s_M$ ; no positivity            |
| Proposition 4 | D1–D3 ( $W_H$ ), D5, D6 (general symbols), D7, D9, D10  | (A1)–(A4), (A6); Proposition 3, Theorem 3                           |
| Corollary 5   | D6 (either set of symbols), D7, D8                      | none of (A1)–(A6): any two laws; Proposition 2; sampling hypotheses |

**Theorem 1 (Indistinguishability criterion).** Under the model of Section 3, with common onset time  $\tau$ , mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  induce identical distributions over the observation trajectory  $\{y_t\}_{t \geq 0}$  if and only if

$$\theta_K C b_K = \theta_F C b_F. \quad (*)$$

No rank, observability or definiteness condition on  $A$ ,  $C$ ,  $Q$  or  $\Sigma$  is needed; the onset  $\tau$  may be any deterministic time.

**Corollary 1 (Fault-mimicking compromise).** Fix  $u = C b_K$ ,  $v = C b_F$ , and a nonzero compromise severity  $\theta_K \neq 0$ . If the fault severity  $\theta_F$  may be any real number, then

$$\exists \theta_F \in \mathbb{R} : \theta_K u = \theta_F v \iff u \in \text{span}(v).$$

Two precisions. Linear dependence of  $u$  and  $v$  means rank at most one, and its subcases differ: if  $v = 0$  but  $u \neq 0$  no  $\theta_F$  solves the equation, and if both vanish every  $\theta_F$  does; with positive severities, nonzero  $u, v$  must lie on the same ray. Laws averaged over onsets would need a model this paper does not specify.

**Corollary 2 (Separation implies distinguishability, not high accuracy).** If  $\text{rank}([Cb_K \ Cb_F]) = 2$ , no pair  $(\theta_K, \theta_F)$  other than  $(0, 0)$  satisfies (\*); for any instance pair with  $(\theta_K, \theta_F) \neq (0, 0)$ ,  $K$  and  $F$  are then distinguishable in the population limit.

Separation does not imply good finite-sample classification: the projected directions can be small relative to the noise. The benchmark is the whitened separation  $d$  of Definition D5; under equal priors the Bayes-optimal accuracy and area under the ROC curve (AUC) are the standard identities [33, 34]

$$\text{Acc}^* = \Phi(d/2), \quad \text{AUC}^* = \Phi(d/\sqrt{2}).$$

**Proposition 1 (Cost of ambiguity is bounded below under confounding).** Suppose (\*) holds (the confounded case),  $0 < p < 1$ ,  $\text{DISRUPT} > 0$ , and  $\text{HARM} > C_{OP}$  (harm strictly worse than routine containment). Then the best achievable expected cost over the policies of Definition D7 (randomized policies allowed) is

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, \ p \text{HARM}\},$$

the oracle's expected cost is  $R_{\text{oracle}} = p C_{OP}$ , and

$$\boxed{\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1 - p) \text{DISRUPT}, \ p(\text{HARM} - C_{OP})\} > 0.}$$

Exact equality is a brittle operational target, so we extend the bound to approximate confounding.

**Proposition 2 (Total-variation extension of Proposition 1).** Let the loss table and prior be as in Proposition 1, with  $0 < p < 1$ ,  $\text{DISRUPT} > 0$  and  $\text{HARM} > C_{OP}$ , but do not assume (\*). Let  $f_K, f_F$  be the observation densities (Definition D8) and set  $a = p(\text{HARM} - C_{OP})$ ,  $b = (1 - p) \text{DISRUPT}$ . Then

$$\text{CoA}^* = \int \min\{af_K(y), bf_F(y)\} dy = \frac{a + b - \int |af_K - bf_F| dy}{2},$$

$$\min(a, b) \left(1 - \text{TV}(P_K, P_F)\right) \leq \text{CoA}^* \leq \min(a, b),$$

and, if a common surrogate law  $\bar{P}$  satisfies  $\text{TV}(P_K, \bar{P}) \leq \epsilon_K$  and  $\text{TV}(P_F, \bar{P}) \leq \epsilon_F$ , then

$$\text{CoA}^* \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}.$$

The surrogate-law bound states the premise any “approximate confounding transfers approximately” argument needs: a bound on distributional approximation error, not a match on one statistic. A classifier performing near chance cannot certify a small total variation distance. The surrogate-law bound is the cost counterpart of  $\epsilon$ -stealth in stochastic control [12]: an attacker whose observation law stays within a small distance of a law a fault can also produce keeps the cost of ambiguity near its maximum.

For the equal-covariance Gaussian setting, likelihood-ratio integration [31] gives the population-optimal ambiguity cost as a function of  $d$ , the unequal-prior cost analog of Corollary 2; a summary-based classifier can only do worse.

**Proposition 3 (Gaussian unequal-prior ambiguity cost).** Let  $Y \mid M \sim \mathcal{N}(\mu_M, V)$ ,  $M \in \{K, F\}$ , with common covariance  $V \succ 0$  and discriminability  $d > 0$  (Definition D5), prior  $p = P(K)$ , and the general loss table with  $a = p(H_K - C_K)$ ,  $b = (1 - p)(C_F - H_F)$ ,  $a, b > 0$ . Then the population-optimal ambiguity cost is

$$\text{CoA}_Y^*(d) = a \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right) + b \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

which tends to  $\min(a, b)$  as  $d \rightarrow 0^+$  (the value at  $d = 0$ , exact confounding, recovering Proposition 1, is obtained as this limit) and decreases toward 0 as  $d \rightarrow \infty$ .

#### 4.1. Confounding is fragile within the linear model: state-level injection

Theorem 1 models the incident as an output-level disturbance: the state recursion does not see it (Definition D2). We also model a compromise or a persistent fault as a state-level disturbance, behaviour at  $t$  entering the state at  $t + 1$ , since an incident’s orders move prices [35] and change the book that other agents react to. This is a modelling assumption, which the simulator study does not test (Section 7.1). This changes the confounding condition. The objects involved are those of geometric failure detection and of attack-space analysis in secure control [10, 36].

**Definition D9 (State-level injection model).** As Definition D2, except that the perturbed state drives the recursion from the onset on:

$$\begin{aligned} x_{t+1} &= Ax_t + w_t \quad (t < \tau), & x_{t+1} &= A\tilde{x}_t + w_t \quad (t \geq \tau), \\ \tilde{x}_t &= x_t + \mathbb{K}[t \geq \tau] \theta_M b_M, & y_t &= C\tilde{x}_t + \varepsilon_t. \end{aligned}$$

The injection is persistent (added at every  $t \geq \tau$ ) and propagates through  $A$ .

**Definition D10 (Unobservable subspace; observability).** The (Kalman) unobservable subspace of  $(A, C)$  is  $\mathcal{N}(A, C) = \bigcap_{k=0}^{n-1} \ker(CA^k)$ . The pair  $(A, C)$  is observable if  $\mathcal{N}(A, C) = \{0\}$ , equivalently if the stacked matrix

$$\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has rank  $n$ .

**Theorem 2 (Confounding under state-level injection requires unobservability).** Under the state-level injection model, let  $\Delta_0 = \theta_K b_K - \theta_F b_F$ . Mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  are observationally indistinguishable at every  $t \geq \tau$  if and only if  $\Delta_0$  lies in the unobservable subspace of the pair  $(A, C)$ , i.e.

$$\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k),$$

the classical (Kalman) unobservable subspace of  $(A, C)$ , where  $n$  is the state dimension.

**Theorem 3 (Finite-horizon version of Theorem 2).** Under the state-level injection model, let  $\Delta_0 = \theta_K b_K - \theta_F b_F$ , let  $H \geq 1$  be the number of post-onset observations, and define

$$G_H = \begin{bmatrix} C \\ C(I + A) \\ \vdots \\ C \sum_{j=0}^{H-1} A^j \end{bmatrix}.$$

Mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  induce the same law of the observations over the window of  $H$  post-onset observations if and only if

$G_H \Delta_0 = 0$ , which holds if and only if  $CA^j \Delta_0 = 0$  for  $j = 0, \dots, H - 1$ . If  $H \geq n$ , this is equivalent to the all-time condition of Theorem 2,  $\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k)$ .

The result is not inherently asymptotic: for  $H \geq n$ , exact equality over  $H$  post-onset observations already forces the all-time Kalman condition. What depends on horizon and noise is discriminability when the condition fails, which Theorem 2 does not bound; Proposition 4 (Section 4.2) gives its population cost.

**Corollary 3 (Confounding under state-level injection with observable dynamics).** Under the hypotheses of Theorem 2, if  $(A, C)$  is observable, then mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$  are observationally indistinguishable at every  $t \geq \tau$  if and only if  $\theta_K b_K = \theta_F b_F$ .

**Definition D11 (Generic).** For fixed  $C$ , a property of  $A$  holds generically if it holds for Lebesgue-almost every  $A \in \mathbb{R}^{n \times n}$ , i.e. it fails only on a Lebesgue-null subset of  $\mathbb{R}^{n \times n}$ .

**Corollary 4 (Theorem 1’s confounded pairs are generically fragile under state-level injection).** Fix  $n \geq 1$  and  $C \in \mathbb{R}^{m \times n}$  with  $C \neq 0$ , and let  $U_C = \{A \in \mathbb{R}^{n \times n} : (A, C) \text{ is not observable}\}$  (Definition D10). Then:

- (i)  $U_C$  is a Lebesgue-null subset of  $\mathbb{R}^{n \times n}$ , so  $(A, C)$  is observable generically (Definition D11);
- (ii) under the hypotheses of Theorem 2, for every  $A \notin U_C$  and every two mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$ , the instances are observationally indistinguishable at every  $t \geq \tau$  if and only if  $\theta_K b_K = \theta_F b_F$ .

The set  $U_C$  depends on  $C$  only, not on the pair. In particular, for every  $A \notin U_C$ , every pair that Theorem 1 confounds with  $\theta_K b_K \neq \theta_F b_F$  is distinguishable under state-level injection.

The pilot’s construction (Section 5), which uses the null space of  $C$  to make  $b_K \neq b_F$  with  $Cb_K = Cb_F$ , is exactly such a pair:  $\Delta_0 \in \ker(C)$  but  $\theta_K b_K \neq \theta_F b_F$ . By Corollary 3 it cannot remain confounded under state-level injection whenever  $(A, C)$  is observable. The pilot’s own  $A$ , a scalar multiple of the identity, is the measure-zero exception, so we re-ran the pilot at both injection levels, with its own  $A$  and with a stable, coupled and observable  $A$  (the *injection-level rerun*), 10 seeds per magnitude, each with 480 balanced test episodes (a constant prediction scores one half). Mean test accuracy over seeds under state-level injection rises from 0.628 at the smallest of 16

magnitudes (95% interval over seeds 0.611–0.645) to 1.000; under output-level injection with the same dynamics it stays between 0.497 and 0.503, and 0 of its intervals exclude one half (Supplement S2). With the pilot’s scalar  $A$ , 1 of the 16 state-level intervals excludes one half (smallest magnitude, 0.501–0.522), about as many as unadjusted intervals on shared seeds give by chance.

A qualification: “almost every” (Corollary 4) concerns a full-measure set of matrices; real, structured market dynamics can have unobservable components by construction, and we have not established that they behave like a generic matrix. Supplement S2 gives three further qualifications.

#### 4.2. Random injections and the observation horizon

Theorems 1–3 compare one fixed instance of each cause, whereas an attacker chooses severity and direction from a class. Theorem 4 lets each cause inject a random vector, drawn once per episode; the instances of Definition D1 are its point-mass case.

**Theorem 4 (Class-level identifiability).** Let each cause  $M \in \{K, F\}$  inject a random vector  $s_M \in \mathbb{R}^n$  in place of  $\theta_M b_M$ , drawn once per episode and persistent (the same draw acts at every  $t \geq \tau$ ), with an arbitrary law, and independent of the pair  $(x_0, (w_t, \varepsilon_t)_{t \geq 0})$  jointly (independence of  $x_0$  and of the noise family separately is not enough) (a mechanism instance of Definition D1 is the point-mass case  $s_M = \theta_M b_M$ ). Write  $\mathcal{O}_H = [C^\top (CA)^\top \cdots (CA^{H-1})^\top]^\top$ .

- (i) Under the output-level injection model (Definition D2), on any window  $W$  that contains at least one  $t \geq \tau$ ,  $K$  and  $F$  induce the same law of  $Y_W$  if and only if  $Cs_K$  and  $Cs_F$  have the same law on  $\mathbb{R}^m$ .
- (ii) Under the state-level injection model (Definition D9), on the window  $W = \{\tau, \dots, \tau + H - 1\}$ , they induce the same law of  $Y_W$  if and only if  $\mathcal{O}_H s_K$  and  $\mathcal{O}_H s_F$  have the same law; for  $H \geq n$ , and for  $W = \{t : t \geq \tau\}$ , if and only if  $\mathcal{O}_n s_K$  and  $\mathcal{O}_n s_F$  have the same law.
- (iii) If  $(A, C)$  is observable (Definition D10), state-level indistinguishability on  $W = \{t : t \geq \tau\}$  holds if and only if  $s_K$  and  $s_F$  have the same law.

**Proposition 4 (Cost of ambiguity over the observation horizon).**

Fix mechanism instances  $(K, \theta_K, b_K)$  and  $(F, \theta_F, b_F)$ ,  $\Delta_0 = \theta_K b_K - \theta_F b_F$ , the general loss table with  $a = p(H_K - C_K) > 0$  and  $b = (1 - p)(C_F - H_F) > 0$ , and for  $H \geq 1$  the window  $W_H = \{\tau, \dots, \tau + H - 1\}$  of the first  $H$  post-onset observations. Let  $V_H$  be the covariance of  $Y_{W_H}$  (it is the same under both

causes and under both injection models), let  $G_H$  be as in Theorem 3, and set

$$d_H^{\text{st}} = \|V_H^{-1/2} G_H \Delta_0\| \quad (\text{state level, Definition D9}), \quad d_H^{\text{out}} = \|V_H^{-1/2} (\mathbf{1}_H \otimes C \Delta_0)\| \quad (\text{output level, De$$

Under either model, with  $d_H$  the corresponding value, the cost of ambiguity on  $W_H$  is  $\text{CoA}_H^* = \text{CoA}_Y^*(d_H)$ , the closed form of Proposition 3, if  $d_H > 0$ , and  $\text{CoA}_H^* = \min(a, b)$  if  $d_H = 0$ . Moreover:

- (i) under either model,  $H \mapsto \text{CoA}_H^*$  is non-increasing;
- (ii) under output-level injection,  $\text{CoA}_H^* = \min(a, b)$  for every  $H$  if  $C\Delta_0 = 0$  (condition  $(*)$  of Theorem 1), and  $\text{CoA}_H^* < \min(a, b)$  for every  $H$  otherwise;
- (iii) under state-level injection,  $\text{CoA}_H^* < \min(a, b)$  if and only if  $CA^j \Delta_0 \neq 0$  for some  $j < H$ ; if  $(A, C)$  is observable and  $\Delta_0 \neq 0$ , this holds for every  $H \geq h^*$ , where  $h^* = 1 + \min\{j : CA^j \Delta_0 \neq 0\} \leq n$ .

In particular, for a pair that Theorem 1 confounds ( $C\Delta_0 = 0$ ) with  $\Delta_0 \neq 0$  and  $(A, C)$  observable, the cost of ambiguity stays at  $\min(a, b)$  at every horizon under output-level injection, and falls strictly below it from horizon  $h^* \in \{2, \dots, n\}$  on under state-level injection.

Figure 1 evaluates Proposition 4 for the pilot's confounded pair.

**From a classifier to the total variation.** The next bound needs no model, only samples from the two laws; Section 7.2 applies it.

**Corollary 5 (Classifier accuracy bounds the total variation and the cost of ambiguity).** Let  $P_K, P_F$  be any two observation laws, as in Proposition 2, with  $a, b > 0$  and  $\text{CoA}^* = \int \min\{af_K, bf_F\} dy$  as there. Let  $g$  be a classifier that maps a feature vector computed from  $Y$  to  $\{K, F\}$ , possibly randomized independently of  $Y$ , with balanced accuracy  $\alpha(g) = \frac{1}{2}[P_K(g = K) + P_F(g = F)]$ . Then

$$\text{TV}(P_K, P_F) \geq 2\alpha(g) - 1, \quad \text{CoA}^* \leq \max(a, b) (1 - \text{TV}(P_K, P_F)) \leq 2 \max(a, b) (1 - \alpha(g)),$$

and every surrogate law  $\bar{P}$  of Proposition 2 has  $\epsilon_K + \epsilon_F \geq 2\alpha(g) - 1$ . Suppose further that  $g$  is fixed given the training data and that, given the training data, the test sample consists of pairs that are independent across pairs, each pair holding one draw from  $P_K$  and one from  $P_F$  (*pipeline unbiasedness*: the generation and evaluation pipeline adds no class-dependent difference of its own). If  $L$  is a lower confidence limit for  $\alpha(g)$  at level  $1 - \delta$  computed from the test sample, then with probability at least  $1 - \delta$  the three bounds hold with  $L$  in place of  $\alpha(g)$ .

Proof: Supplement S1 [30].

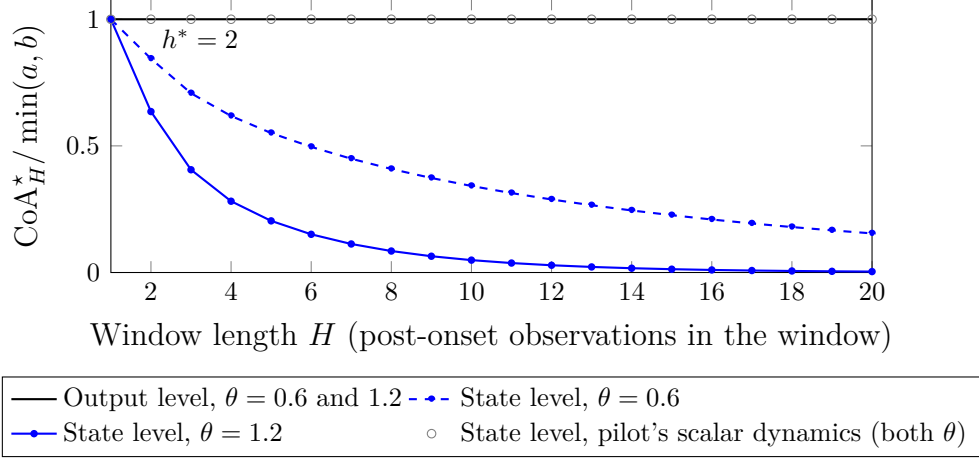


Figure 1: Population cost of ambiguity on the window of the first  $H$  post-onset observations (Proposition 4), as a fraction of  $\min(a, b)$ , for the pilot’s confounded pair ( $Cb_K = Cb_F$ ) at prior  $p = 0.5$ . The pilot’s  $C$ ,  $Q$ ,  $\Sigma$ ,  $\tau$  and injection directions are used with the coupled observable  $A$  of the injection-level rerun (Section 4.1), not with the pilot’s own dynamics. Under output-level injection the cost stays at  $\min(a, b)$  at both magnitudes  $\theta = 0.6$  and  $\theta = 1.2$ ; under state-level injection it falls below  $\min(a, b)$  from  $h^* = 2 \leq n = 3$  on. With the pilot’s own scalar dynamics, which are not observable with the pilot’s  $C$  (observability rank  $2 < n$ ), the state-level cost also stays at  $\min(a, b)$  at every  $H$  (grey markers on the output-level line), the exception of Section 4.1.

## 5. Pilot-scale validation in the linear model

We instantiate the model at small scale (a few latent dimensions, fewer observed channels, AR(1) dynamics; parameters in Supplement S2) as a check that the theory is internally consistent and detectable in simulation, not as a market-realistic study. Two regimes are built by the choice of injection directions: *confounded*, with  $b_K, b_F$  chosen through the null space of  $C$  so that  $Cb_K = Cb_F$  exactly, satisfying Theorem 1’s condition; and *separated*, with  $Cb_K, Cb_F$  linearly independent. A logistic-regression classifier is trained on whole-episode summary statistics of each observed channel (mean, standard deviation, net drift), in both regimes and at each injection magnitude.

In the injection-level rerun (Section 4.1), with the pilot’s own  $A$  and output-level injection, over 10 seeds at each of 16 magnitudes from 0.2 to 2.5, the confounded construction has mean accuracy 0.505–0.512, and 0 of its intervals over seeds exclude one half: as Theorem 1 predicts, a more severe incident is not easier to classify when the two observation laws are identical.

One archived run at two magnitudes (one seed and split, 480 test episodes per cell; Supplement S2) agrees, and in its separated regime accuracy is high and rises with magnitude, as the benchmark  $\text{Acc}^* = \Phi(d/2)$  predicts, since  $d$  grows with the magnitude. Table 2 lists the pilot’s three separate runs: the injection-level rerun, the archived run and the cost pilot of Section 6.

Table 2: The linear pilot’s runs: dynamics, injection level, magnitudes, seeds, test episodes per seed, and the section using each.

| Run                   | Dynamics $A$         | Injection level  | Magnitudes (range) | Seeds | Test episodes per seed | Section |
|-----------------------|----------------------|------------------|--------------------|-------|------------------------|---------|
| Injection-level rerun | Pilot’s own (scalar) | Output and state | 16 (0.2–2.5)       | 10    | 480                    | 4.1, 5  |
| Injection-level rerun | Coupled, observable  | Output and state | 16 (0.2–2.5)       | 10    | 480                    | 4.1     |
| Archived run          | Pilot’s own (scalar) | Output           | 2 (0.6–1.2)        | 1     | 480                    | 5       |
| Cost pilot            | Pilot’s own (scalar) | Output           | 1 (1)              | 20    | 480                    | 6       |

Three further classifier families (a random forest, Gaussian naive Bayes and a single-feature threshold rule) give the same pattern (Supplement S2), as Theorem 1 implies for any passive classifier.

## 6. The cost of ambiguity in the linear pilot

**Terminology.** With the oracle and best observation-based risks  $R_{\text{oracle}}$ ,  $R_Y^*$  of Definition D7, the best constant action  $R_{\text{const}} = \min_a \mathbb{E}[L(a, M)]$  and the best risk  $R_Z^*$  of a policy of engineered features  $Z = \phi(Y)$ ,

$$R_{\text{oracle}} \leq R_Y^* \leq R_Z^* \leq R_{\text{const}}.$$

The empirical sections estimate  $R_Z^*$ , equal to  $R_Y^*$  only for sufficient  $\phi$  (not established here). Exact confounding is sufficient, not necessary, for  $R_Y^* = R_{\text{const}}$ : an informative  $Y$  can still leave the optimal action unchanged, for example at a low prior where every posterior stays below the containment threshold. An absence of measured cost benefit is therefore not evidence that two observation laws are equal.

**Prior-shift correction.** The classifier is trained at a balanced prior, so its score is corrected to the deployment prior by Bayes’ rule before it is compared with  $q^* = \text{DISRUPT}/(\text{HARM} - C_{OP} + \text{DISRUPT})$  [37]; without the correction a classifier under confounding can look worse than ignoring the observation (Supplement S4).

**Linear pilot.** With the illustrative toy triple  $(C_{OP}, \text{DISRUPT}, \text{HARM}) = (3, 6, 10)$ , which satisfies Proposition 1’s hypotheses, we compare four policies

Table 3: Linear pilot, confounded regime: expected cost per episode (toy units). Under exact confounding  $R_Y^* = R_{\text{const}}$ , so the last column is the classifier’s excess over the best achievable cost: mean over 20 seeds, each with 480 test episodes, with the range over seeds in brackets. Per-seed values are test-set estimates and can fall on either side of the floor.

| $P(K)$ | Contain | Leave | Classifier (sd) | Oracle | $R_{\text{const}}$ | Classifier $- R_{\text{const}}$ [min, max] |
|--------|---------|-------|-----------------|--------|--------------------|--------------------------------------------|
| 0.05   | 5.850   | 0.500 | 0.500 (0.000)   | 0.150  | 0.500              | 0.000 [0.000, 0.000]                       |
| 0.1    | 5.700   | 1.000 | 1.000 (0.000)   | 0.300  | 1.000              | 0.000 [0.000, 0.000]                       |
| 0.2    | 5.400   | 2.000 | 2.000 (0.000)   | 0.600  | 2.000              | 0.000 [0.000, 0.000]                       |
| 0.3    | 5.100   | 3.000 | 3.000 (0.000)   | 0.900  | 3.000              | 0.000 [0.000, 0.000]                       |
| 0.5    | 4.500   | 5.000 | 4.520 (0.085)   | 1.500  | 4.500              | 0.020 [-0.125, 0.196]                      |

across a grid of priors  $P(K)$ : always contain, always leave, the prior-corrected classifier, and the oracle.

The constant and oracle policies match their closed forms  $pC_{OP} + (1 - p)\text{DISRUPT}$ ,  $p\text{HARM}$  and  $pC_{OP}$ , a direct check of the cost computation. Under confounding no policy of  $Y$  has expected cost below  $R_Y^* = R_{\text{const}}$ , so the fitted classifier can at best reach this floor. It did so exactly at every prior below  $q^* = 0.462$ , taking the best constant action on every test episode in all 20 seeds; at the highest prior, the only one above  $q^*$ , its cost scatters around the floor (Supplement S2). Proposition 1 concerns the Bayes-optimal policy; it implies that a fitted classifier cannot beat the floor, not that it attains it. Under separation the classifier beats the best constant policy (Supplement S2, with a sensitivity analysis over cost triples).

## 7. Observation matching in an agent-based market

### 7.1. Scope and constructions

**What this section does and does not test.** Theorem 1 concerns equality of the whole observation-trajectory law,  $P_K(Y_{0:T}) = P_F(Y_{0:T})$ . The constructions below match the two mechanisms on one chosen summary statistic, the design’s calibration target, while other observable channels may still differ. Matching one moment is not equality of the joint law, and a classifier exploiting an unmatched channel does not contradict Theorem 1, even in a linear system:  $\mathcal{N}((0, 2), I)$  and  $\mathcal{N}((0, -2), I)$  share their first-coordinate mean. No linearization bridge (operating point, Jacobian, approximation-error bound) connects the linear model to the simulator. This section is therefore not a test of Theorem 1; it asks what a one-parameter calibration

toward one summary statistic leaves for a classifier to exploit, and what that does to decision cost.

We inject compromise and fault mechanisms as order attributes (submission rate, order size, price aggressiveness, directional bias, cancel/replace churn) into ABIDES [38], a discrete-event multi-agent market simulator with a background population of noise, value and market-maker agents. Each design has two regimes. In the *summary-matched* regime (named for its target, not for a match achieved; Section 7.2) a calibration search scales the fault’s attributes by one magnitude  $\theta_F$ , moving each from its pre-onset baseline  $z_0$  (one for a multiplier, zero otherwise) to  $z_0 + \theta_F(z - z_0)$ , with  $z$  its listed value, with the compromise fixed, toward the compromise’s value of a chosen target statistic (the direction of the search is not symmetric; Section 7.2); in the *contrast* regime the two mechanisms are deliberately given a large difference in a chosen attribute, which the observed channels need not expose: it is not a positive control. The classifier (logistic regression) uses whole-episode summaries of the observed channels (Section 3.5).

**Four designs, five profiles.** Table 4 defines designs *D1–D4*: varied attributes, summary-matched attribute values, calibration target, chosen  $\theta_F$  and observed channels. *D2*’s incident agent also submits on a fixed clock, not at Poisson times, with a fixed order size. Profile *D1+* reuses *D1*’s episodes with three added channels, among them order-flow imbalance [35]; it is a richer feature set on the same data, not a replication, so 50-cell counts include those episodes twice. The search evaluates the grid  $\theta_F \in \{0.5, 1, 1.5, 2\}$  on a few dozen episodes each and keeps the value whose mean target is closest to the compromise’s, with no tolerance. All designs share one background population, session length and onset (Supplement S3).

**Era-3 basis.** Every headline ABIDES result is generated with the third, hardened version of the simulation pipeline (the *Era-3* pipeline: separate worker processes, one task per worker, and a runtime check that the random-number-generator patches are active; Section 7.4, Supplement S6), on *paired* seeds, each yielding one compromise and one fault episode; earlier-era results are exploratory (Section 7.4).

## 7.2. Classification on the paired basis

Every cell has 1000 episodes per class and 600 held-out test episodes from 300 test seeds. A cell is *at chance* when its accuracy interval contains one half, *above chance* when the lower end of its accuracy interval is above one half, and *beyond the control* when that lower end also exceeds the upper

Table 4: The designs: varied attributes, summary-matched compromise ( $K$ ) and fault ( $F$ ) attributes, calibration target, the searched fault magnitude  $\theta_F$  at its chosen value (the compromise is fixed), and observed channels. Listed values  $z$ ; the fault’s scaled values  $z_0 + \theta_F(z - z_0)$ , which generated the cells, are in brackets (Supplement S3).

| Design | Varied attributes                                                       | Summary-matched attributes                                                                                                                                                                                       | Calibration target           | Searched parameter (chosen) | Observed channels                                                                                             |
|--------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| D1     | buy bias, size multiplier, rate multiplier, aggressiveness, cancel rate | $K$ : buy bias 0.8, size multiplier 1.5, rate multiplier 1.5, aggressiveness 3, cancel rate 0.05 ( $\theta_K = 1$ ); $F$ : buy bias 0.8, size multiplier 3, rate multiplier 3, aggressiveness 0, cancel rate 0.7 | mid-price drift              | $\theta_F = 1$              | spread, depth, cancel rate, mid price                                                                         |
| D1+    | buy bias, size multiplier, rate multiplier, aggressiveness, cancel rate | $K$ : buy bias 0.8, size multiplier 1.5, rate multiplier 1.5, aggressiveness 3, cancel rate 0.05 ( $\theta_K = 1$ ); $F$ : buy bias 0.8, size multiplier 3, rate multiplier 3, aggressiveness 0, cancel rate 0.7 | mid-price drift              | $\theta_F = 1$              | spread, depth, cancel rate, mid price, order flow imbalance book, signed executed volume, queue concentration |
| D2     | buy bias, aggressiveness, cancel rate                                   | $K$ : buy bias 0.85, aggressiveness 1.2, cancel rate 0.05 ( $\theta_K = 1$ ); $F$ : buy bias 0.85, aggressiveness 0.1, cancel rate 0.8                                                                           | mean cancel rate             | $\theta_F = 1$              | spread, depth, cancel rate, mid price                                                                         |
| D3     | size multiplier, rate multiplier, cancel rate                           | $K$ : size multiplier 1.8, rate multiplier 1.8, cancel rate 0.1 ( $\theta_K = 1$ ); $F$ : size multiplier 1 [1], rate multiplier 2.5 [1.75], cancel rate 0.5 [0.25]                                              | total post-onset order count | $\theta_F = 0.5$            | spread, depth, cancel rate, mid price                                                                         |
| D4     | buy bias, size multiplier, aggressiveness                               | $K$ : buy bias 0.8, size multiplier 1.5, aggressiveness 2 ( $\theta_K = 1$ ); $F$ : buy bias 0 [0], size multiplier 4 [5.5], aggressiveness 0 [0]                                                                | mean spread                  | $\theta_F = 1.5$            | spread, depth, cancel rate, mid price                                                                         |

Table 5: Classification by design and regime. The ten classification cells are computed on paired seeds: each seed yields one compromise and one fault episode, the train/test split is grouped by seed, and the intervals are cluster-bootstrap intervals over seeds, all under the hardened pipeline (Section 7.4). <sup>†</sup> Zero test errors, so the bootstrap interval is degenerate: accuracy has the exact Clopper–Pearson interval over test seeds (a seed counts as correct when both its episodes are), and AUC the lower bound  $\text{TPR}_{\text{lo}} \cdot \text{TNR}_{\text{lo}}$ , the product of exact one-sided Clopper–Pearson lower bounds at 98.75% on each rate (joint level 97.5%, the one-sided level of the lower end of a two-sided 95% interval), valid because  $\text{AUC} \geq \text{TPR} \cdot \text{TNR}$  at the fixed threshold (Supplement S6). Other bounds shown as one are rounded. Column  $2L - 1$ : Corollary 5’s lower bound on the total variation between the two laws, with  $L$  the lower end of the accuracy interval.

| Design | Accuracy [95% CI]    | Contrast                | $2L - 1$ | Summary-matched                   |                                   |          |
|--------|----------------------|-------------------------|----------|-----------------------------------|-----------------------------------|----------|
|        |                      | AUC [95% CI]            |          | Accuracy [95% CI]                 | AUC [95% CI]                      | $2L - 1$ |
| D1     | 0.685 [0.652, 0.718] | 0.756 [0.728, 0.786]    | 0.303    | 0.857 [0.830, 0.882]              | 0.934 [0.917, 0.950]              | 0.660    |
| D1+    | 0.998 [0.995, 1.000] | 1.0000 [0.9999, 1.0000] | 0.990    | 0.898 [0.875, 0.920]              | 0.966 [0.954, 0.977]              | 0.750    |
| D2     | 0.663 [0.633, 0.692] | 0.731 [0.706, 0.757]    | 0.267    | 0.553 [0.525, 0.580]              | 0.577 [0.551, 0.604]              | 0.050    |
| D3     | 0.998 [0.995, 1.000] | 1.0000 [0.9998, 1.0000] | 0.990    | 0.875 [0.850, 0.898]              | 0.953 [0.940, 0.967]              | 0.700    |
| D4     | 0.727 [0.698, 0.757] | 0.807 [0.783, 0.832]    | 0.397    | 1.000 [0.988, 1.000] <sup>†</sup> | 1.000 [0.971, 1.000] <sup>†</sup> | 0.976    |

end of the negative control’s accuracy interval, the pipeline bias the control cannot exclude (Section 7.4); the interval is the cluster bootstrap over test seeds, or the exact Clopper–Pearson interval for a zero-error cell. Read this way, each cell is a classifier two-sample test [32]: accuracy above one half rejects equality of the two mechanisms’ observation laws if the pipeline is unbiased (Section 7.4). The test is symmetric in the labels, so each summary-matched cell above chance shows that its pair is distinguishable, while a compromise tuned to a fixed fault is untested. Under the same condition, Corollary 5 turns the lower end  $L$  of a cell’s accuracy interval into  $\text{TV} \geq 2L - 1$  (Table 5). In D1, D1+, D3 and D4 no law then lies close to both mechanisms’ laws: Proposition 2’s surrogate-law floor still holds, but is at most  $2(1 - L) \min(a, b)$ , far from  $\min(a, b)$ . In D2 the bound is within what a pipeline bias the control leaves open could produce.

Table 5 shows no single ranking. In D1 the summary-matched regime is the more classifiable one, with non-overlapping accuracy intervals; with D1+’s added channels the contrast regime becomes nearly perfectly classifiable, because the contrast mechanisms were always separable on signed executed volume, which the original four channels did not expose (Supplement S3). D2 is weakly classifiable in both regimes; for it and for D1’s summary-matched regime no single leaking channel was identified (the best single-statistic rule of Supplement S4 is descriptive evidence of the dominant channel). In D3 both regimes are strongly classifiable: its mechanisms differ in cancellation rate, one of the observed channels, which matching total order count left free. D4’s summary-matched cell has 0 test errors (exact 95% Clopper–Pearson lower bound 0.988 over test seeds): calibrating on mean spread left a large difference in effective order size, which leaks through depth: a size-based channel, not the cancellation-rate channel of D3. Matching also failed on the target itself: a threshold on mean spread alone beats the best constant policy with a supported interval at every prior but the lowest (Supplement S4), while depth remains the dominant leak. In each of the four injection designs, calibration by search left the summary-matched regime above chance (Table 5), and beyond the control in D1, D3 and D4. In D2 the accuracy interval (0.525–0.580) and the AUC interval (0.551–0.604) start close to one half, below the control’s upper end: D2 is above chance but not beyond the control, so it is indeterminate as evidence that its pair’s observation laws differ. These are per-cell intervals, not adjusted for the 10 comparisons. With max-T simultaneous intervals over the 10 cells (Supplement S4), D2’s summary-matched accuracy interval is 0.512–0.594 and its

AUC interval 0.535–0.619, still above one half: every summary-matched cell stays above chance, and D2 stays short of the control.

**What the search did.** It kept the fault at its listed values,  $\theta_F = 1$  in D1 and 1 in D2, and scaled it only in D3 and D4 (Table 4). No construction was matched to a pre-specified tolerance on independent data, so “summary-matched” means a match attempted, not achieved: D1’s selected point overshoots its target in every confirmation run (Supplement S3), and D3’s target is not a stored feature. In D1, D2 and D4 the achieved difference on the target, compromise minus fault over all paired episodes, has a cluster-bootstrap interval that excludes zero (Supplement S3): the search matched the target in none of the designs where it can be measured. The cells therefore do not separate leakage through unmatched channels from a residual difference on the target.

### 7.3. Cost of ambiguity on the paired basis

**The dollar cost triple.** For the ABIDES study we reconstruct a dollar cost triple  $(C_K, C_F, H_K) = (C_{OP}, \text{DISRUPT}, \text{HARM})$  from the Almgren–Chriss market-impact model [39] in its continuous-trading form, the expected shortfall of a uniform-rate liquidation,  $L(X, T) = \varepsilon X + (\frac{\gamma}{2} + \frac{\eta}{T})X^2$ ; the discrete-interval correction it drops, and its size for each window, are in Supplement S4. The triple uses the model’s own illustrative parameters applied to three scenarios: a routine containment trade, a benign order forced into an urgent unwind, and an unmitigated adversarial sequence. This gives  $(C_K, C_F, H_K) = (\$490,938, \$473,958, \$1,317,500)$ , which satisfies  $H_K > C_K$ . It is an arithmetic illustration, not a calibration to any venue or incident (worked example in Supplement S4). Every policy comparison below depends on the costs only through  $a/b$ , with  $a = p(H_K - C_K)$  and  $b = (1-p)(C_F - H_F)$ : scaling both leaves the prior-corrected classifier’s actions and the sign of every cost difference unchanged. The prior grid is therefore also a grid of cost ratios at a fixed prior, and every verdict below holds for any loss table with the same per-cell ratio  $(H_K - C_K) : (C_F - H_F)$ . Supplement S4 reads the classifier against the best constant at point estimate over 49 priors from 0.02 to 0.98, that is over a wide range of cost ratios.

**A positive fault loss.** The simulated fault mechanisms submit real orders, so leaving a fault alone is not free. We replace the zero entry by  $H_F$ , the Almgren–Chriss cost of the fault mechanism’s directly measured submitted volume in each episode, averaged over episodes (so the convexity of  $L$  in  $X$  is included); it is a modelled cost of submitted, not executed, volume,

measured on the same episodes as the classification cells (Section 7.4). Per-design values are in Supplement S4; they range from \$850 to \$104,739 per episode. In D2 the fault’s submitted volume is deterministic (fixed clock and order size, Section 7.1), so  $H_F$  and the rebaselined  $C_F$  are the same in both regimes (Supplement S4). With  $H_F > 0$ ,

$$R_{\text{oracle}} = p C_K + (1 - p) H_F,$$

$$R(\pi) - R_{\text{oracle}} = p (H_K - C_K) \text{FNR} + (1 - p) (C_F - H_F) \text{FPR},$$

and the cost-optimal threshold on the prior-corrected score is the  $q^*$  of (A7), which depends on the loss table only (between 0.309 and 0.364 under the current convention and between 0.004 and 0.379 under the rebaselined one; per unit in Supplement S4). In the general table Proposition 1’s floor is  $\min(a, b)$ , the exact-confounding value of Propositions 3–4. Each cell’s cost is estimated by direct class-conditional weighting,  $\hat{R}_p = p\{C_K + (H_K - C_K)\hat{\text{FNR}}\} + (1 - p)\{H_F + (C_F - H_F)\hat{\text{FPR}}\}$ , from the held-out error rates at  $q^*$ ; this point estimate uses no resampling, and its intervals are cluster-bootstrap intervals over test seeds.

**Two fault-cost conventions.** Both conventions share the compromise column,  $C_K = L(X_0, T_u)$  and  $H_K$  of the dollar triple, with  $X_0$  its reference order and  $T_u$  its urgent horizon, and the same  $H_F = \frac{1}{N} \sum_i L(X_i, T_n)$ : the cost of each fault episode’s measured submitted volume  $X_i$  at the pace it was submitted (the post-onset window,  $T_n$ ). They differ only in  $C_F$ .

- *Current convention (primary).*  $C_F = \text{DISRUPT} = L(X_0, T_u) - L(X_0, T_p)$ , with  $T_p$  the patient horizon: the premium of forcing the reference benign order into an urgent rather than a patient unwind, one value for every cell. It models containment that stops the fault’s orders before they execute but disrupts benign business; like  $H_F$ , it is an excess cost over the incident-free counterfactual, in which the benign order runs patiently and the fault’s orders do not exist.
- *Rebaselined convention (sensitivity arm).*  $C_F = \frac{1}{N} \sum_i L(X_i, T_u)$ : containing a fault means unwinding urgently the volume the fault has already put on. Both fault entries are then absolute costs of the same volume, differing only in urgency, so  $C_F - H_F$  is an urgency premium, positive because the urgent horizon  $T_u$  is shorter than the fixed fault horizon  $T_n$ ;  $C_F$  varies by design and regime (Supplement S4).

Both conventions satisfy (A7),  $H_K > C_K$  and  $C_F > H_F$ , in every cell (minimum  $C_F - H_F$ : 369,219 current, 2,914 rebaselined); the formulas above require it. Which convention fits depends on whether containment precedes the fault’s executions: a modelling choice, not a data question. We report the current convention as primary, as fixed in the analysis plan before the table was computed, and the rebaselined convention as a sensitivity arm.

The cost table applies this triple and  $H_F$  to five profiles  $\times$  two regimes  $\times$  five priors (50 cells; full table in Supplement S4). Intervals on differences between policies are cluster-bootstrap intervals over test seeds; the constant policies’ costs are not resampled.

Against the best constant policy, the classifier’s advantage is a supported improvement in 40 of 50 cells, an exact tie in 2 and inconclusive at this power in 8; at point estimate (strict inequality) it is lower in 44. No cell shows a supported deterioration; no non-inferiority margin was specified, so this is not a non-inferiority result. Always-contain, the action of a containment-first kill switch, is the best constant only at the highest prior under this convention; the classifier beats it in 50 cells, with a supported improvement in 49. The 50 cells are not 50 independent comparisons: the five priors of a profile reweight the same test episodes, D1+ re-featurizes D1’s episodes, and the two regimes of a design share seeds, so the cells come from 10 profile–regime units drawn from 8 episode sets. Each verdict uses its own interval, not adjusted for the number of cells. With max-T simultaneous intervals over each family of cells (Supplement S4), the improvement over the best constant is supported in 39 cells under the current convention and 32 under the rebaselined one, and every unit keeps its class in Table 6. By unit (Table 6), the improvement over the best constant is supported at every prior in 6 units, at some priors in 3, and at none in 1, the D2 summary-matched unit. The cells without a supported improvement (Table 7) lie at the low end of the prior grid or in D2; in all of them the best constant policy is always-leave, except the D2 summary-matched cell at the highest prior, where it is always-contain.

The exceptions are concentrated in D2 (5 of the 6 point-estimate exceptions); the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 7). At the lowest priors of D2’s summary-matched regime the classifier’s cost equals the best constant’s exactly: a tie, counted as an exception under strict inequality. Under the rebaselined fault-cost convention always-contain is the cheaper constant action in 36 of 50 cells:  $q^*$  lies below the whole prior grid in both regimes of D2 and D3 and in D4’s contrast regime, so there it

Table 6: Cost verdicts against the best constant policy by profile–regime unit, under both fault-cost conventions: priors (of five) with a supported improvement; exceptions (priors where the classifier’s point estimate does not beat it); and the best constant at each prior, lowest first (L: always-leave; C: always-contain). Per-cell cluster-bootstrap intervals, unadjusted for multiplicity; all cells in Supplement S4.

| Design | Regime          | Current convention |            |               | Rebaselined convention |            |               |
|--------|-----------------|--------------------|------------|---------------|------------------------|------------|---------------|
|        |                 | Supported          | Exceptions | Best constant | Supported              | Exceptions | Best constant |
| D1     | contrast        | 3                  | 1          | LLLLC         | 3                      | 0          | LLCCC         |
| D1     | summary-matched | 5                  | 0          | LLLLC         | 5                      | 0          | LLLLC         |
| D1+    | contrast        | 5                  | 0          | LLLLC         | 5                      | 0          | LLCCC         |
| D1+    | summary-matched | 5                  | 0          | LLLLC         | 5                      | 0          | LLLLC         |
| D2     | contrast        | 3                  | 1          | LLLLC         | 0                      | 5          | CCCCC         |
| D2     | summary-matched | 0                  | 4          | LLLLC         | 0                      | 5          | CCCCC         |
| D3     | contrast        | 5                  | 0          | LLLLC         | 5                      | 0          | CCCCC         |
| D3     | summary-matched | 5                  | 0          | LLLLC         | 3                      | 1          | CCCCC         |
| D4     | contrast        | 4                  | 0          | LLLLC         | 3                      | 0          | CCCCC         |
| D4     | summary-matched | 5                  | 0          | LLLLC         | 5                      | 0          | LLCCC         |

Table 7: Cells whose advantage over the best constant policy is not a supported improvement (paired basis, current fault-cost convention). Costs in dollars; difference = classifier minus best constant, with its cluster-bootstrap interval over seeds. Tie: interval zero at both ends.

| Design | Regime          | Prior | $R_{\text{clf}}$ | Best constant     | Difference [95% CI]    | vs best constant | vs always-contain |
|--------|-----------------|-------|------------------|-------------------|------------------------|------------------|-------------------|
| D1     | contrast        | 0.05  | 99,906           | 97,510 (leave)    | 2,395 [−1,791, 7,564]  | inconclusive     | improvement       |
| D1     | contrast        | 0.10  | 160,009          | 161,720 (leave)   | −1,711 [−7,827, 5,451] | inconclusive     | improvement       |
| D2     | contrast        | 0.05  | 67,767           | 66,682 (leave)    | 1,085 [−827, 4,357]    | inconclusive     | improvement       |
| D2     | contrast        | 0.10  | 128,699          | 132,515 (leave)   | −3,816 [−6,894, 401]   | inconclusive     | improvement       |
| D2     | summary-matched | 0.05  | 66,682           | 66,682 (leave)    | 0 [0, 0]               | tie              | improvement       |
| D2     | summary-matched | 0.10  | 132,515          | 132,515 (leave)   | 0 [0, 0]               | tie              | improvement       |
| D2     | summary-matched | 0.20  | 266,863          | 264,180 (leave)   | 2,683 [−551, 7,178]    | inconclusive     | improvement       |
| D2     | summary-matched | 0.30  | 400,582          | 395,845 (leave)   | 4,737 [−6,297, 15,234] | inconclusive     | improvement       |
| D2     | summary-matched | 0.50  | 481,261          | 482,448 (contain) | −1,187 [−6,308, 4,133] | inconclusive     | inconclusive      |
| D4     | contrast        | 0.05  | 69,353           | 71,486 (leave)    | −2,133 [−5,648, 2,486] | inconclusive     | improvement       |

is the best constant at every prior. Against the best constant policy, over all 50 cells, the classifier’s improvement is supported in 34, an exact tie in 10 and inconclusive in 6, and none is a supported deterioration against either comparator. It is worse at point estimate in 1 cell, in D3; in the 10 D2 cells it contains every test episode, so it ties exactly with always-contain. The cells without a supported improvement are no longer confined to the low end of the prior grid and D2 (Table 6), so the location statement above holds under the current convention only (Supplement S4).

**Practice baseline.** Against the best rule thresholding one stored episode statistic, fitted on the training seeds (Supplement S4), the classifier shows no supported deterioration under the current convention; under the rebaselined convention 1 cell is a supported deterioration (D1, contrast regime, prior 0.30; the best rule there thresholds mean spread), inconclusive after a max-T multiplicity adjustment (Supplement S4). There the classifier’s expected cost is 261,875 against the rule’s 250,231 dollars, a difference of 11,644 (per-cell interval from 3,961 to 19,982; max-T interval from  $-1,995$  to 25,282). In the summary-matched regime the classifier improves on the rule with a supported interval in D1, D1+ and D3, and is inconclusive in D2; in D4, where it is at ceiling, a depth-variability threshold takes the same actions, so the classifier adds nothing there.

#### 7.4. Checks that bound the claims

**Negative control.** Two incident variants with identical order-generation kernels (the same distribution of rate, size, side, aggressiveness and cancellation, differing only in an inert label), run through the full pipeline in D1’s configuration in 20 independent replicates, each of 50 unpaired episodes per class with 15 per class held out (600 test episodes in all), give mean accuracy 0.505 with a 95% interval over replicate means 0.466–0.544, inside the pre-specified equivalence margin 0.45–0.55; the AUC interval (0.444–0.522) is not, so AUC is inconclusive at this power. Section 7.2 reads D2 against it (Supplement S5). Smaller than the cells, unpaired and run in D1’s configuration only, the control gives a rough bias bound set by its own sampling error, not a measured bias: a larger control with the same mean could move D2 beyond it, not D1, D3 or D4 below it.

**Provenance of each result.** The ten headline classification cells (Table 5), the cost table built on them and  $H_F$  were generated with the hardened pipeline on paired seeds;  $H_F$  is computed per fault episode from those episodes (Supplement S4). The multiplicity adjustment, the practice baseline

and leave-one-design-out (Supplement S5) reanalyse these episodes; the negative control is also hardened, with registered seed blocks. The persistence ablation (an incident shift that persists across wake-ups versus one that fires once; Supplement S5) and the secondary studies of Supplement S3, except its matching-tolerance estimates and thirty-split check, come from the two earlier generation eras; they are reported as exploratory with their era; their seed blocks were registered by a later audit, some generation dates unknown (Supplement S6).

#### *7.5. Outlook: an action on a second information channel*

In the cells of Table 7 (Section 7.3), the fitted summary-feature classifier’s advantage over the best constant policy is not a supported improvement. This bounds  $R_Z^*$ , not  $R_Y^*$  (Section 6): it does not show that passive observation adds no value there. Whether an action on a second information channel, such as targeted verification, adds value in those cells is open, and this paper claims no answer; in particular, it does not claim that these cells are where such an action’s effect is largest. The question is addressed, as an exploratory test, in a working paper [40].

This paper defines no confounded region for a real incident and evaluates no verification channel: confounding (Theorem 1) is a property of the mechanism instances and the observation law, not of the prior  $p$ , and only the value of resolving it, Proposition 1’s  $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$ , depends on  $p$ .

## **8. Discussion**

The two studies bear unequally on one premise, that a fault and a compromise can be made indistinguishable. In the linear model it needs equal laws, which state-level injection generically breaks. In the simulator, a one-parameter calibration toward one statistic left the laws apart, through unmatched channels or a residual difference on the target, which the cells do not separate; they refute Proposition 2’s premise for three constructions. A third caution is methodological: a threshold set without prior-shift correction can manufacture a cost finding (Supplement S4). Supplement S7 turns these observations into a practitioner checklist.

## 9. Limitations

1. **Retrospective, whole-episode diagnosis.** No result measures the value of acting at a decision deadline (Section 3.5).
2. **One simulator, one shared infrastructure.** The four ABIDES designs are parameterizations of one shared code base (configuration builder, kernel, features, cost pipeline), not independently coded implementations, so a bug common to that infrastructure would not be caught by comparing designs. The five profiles are regularities of the constructions tested; the design space is open-ended. Independent evidence comes from four designs; the 10 profile-regime units of Section 7.3 are not independent.
3. **Illustrative costs.** The dollar triple is an illustration, not a calibration (Section 7.3), and  $H_F$  prices submitted, not executed, volume with a market-impact model; neither is a measured economic loss.
4. **Reference policies.** The constant policies are decision-theoretic reference points, not models of deployed kill switches; the threshold-rule baseline uses stored episode statistics only, so an operational comparison would still need a detector with comparable feature access and timing.
5. **Power and degenerate cells.** The cells of Table 7 other than its ties are inconclusive at the sample size used, not shown to be ties. The persistence ablation is exploratory (Section 7.4).

## 10. Conclusion

In a linear-Gaussian model of gateway-observable order flow, confounding of a compromise and a fault is generically fragile: under state-level injection it survives only on the Kalman unobservable subspace (Section 4.1). Theorem 4 extends this to random injections, and Proposition 4 to the cost of ambiguity over the observation horizon.

In the ABIDES simulator, calibration searches that tuned the fault toward a fixed compromise, without reaching a pre-specified match, left the summary-matched regime above chance in all four designs, also after multiplicity adjustment, and beyond the rough bias bound of a small negative control in D1, D3 and D4; D2 is indeterminate.

At one illustrative cost triple, the classifier-based policy improves on the best constant policy with a supported interval in 40 of 50 cells (39 after max-T adjustment), and no cell is supported worse, which is not non-inferiority;

in D2, under the rebaselined convention, it only reproduces always-contain (Section 7.3); Section 7.5 states what the remaining cells leave open.

### **Data and code availability**

The simulation code, the experiment configurations and the result file behind every number in this article are in the project’s public repository, <https://github.com/mejrissi/Thesis>. It holds the provenance register (`experiments/provenance_register.csv`), the validation report (`docs/validation_report.md`), the results files in `results/` with their manifests, and the patches applied to the open-source ABIDES market simulator (`external/patches/`). The register lists each result’s seed block and generation era (Section 7.4, Supplement S6). A tagged release of the repository is made at submission, and an archival DOI is added on acceptance.

### **CRedit authorship contribution statement**

**Chiheb Mejrissi:** Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

### **Declaration of competing interest**

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### **Funding**

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

### **Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the author used Claude (Anthropic), a generative AI assistant, to draft and edit text, to write and check analysis and table-building code, and to check derivations and internal consistency. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

## Appendix A. Proofs of the main results

This appendix proves Theorem 4, Corollary 4 and Proposition 4 from main-text statements only. The last paragraph of the first proof derives Theorems 1–3 and Corollary 3 as the point-mass case. Supplement S1 proves every numbered statement separately.

### Appendix A.1. Proof of Theorem 4

*Proof. Step 0 (decomposition).* Let  $y_t^0 = Cx_t^0 + \varepsilon_t$  be the observation the model produces with no injection, driven by the same  $x_0$  and noise;  $x_{t+1}^0 = Ax_t^0 + w_t$ . By (A1)–(A3), with (A2) in the form the proofs use ( $x_0$  Gaussian, possibly degenerate, independent of the noise, with an instance-independent law), for every finite window  $W$  the vector  $Z_W = (y_t^0)_{t \in W}$  is Gaussian,  $Z_W \sim \mathcal{N}(\mu_W, V_W)$  with  $V_W$  positive semidefinite, and its law does not depend on the cause. Since  $Z_W$  is a measurable function of the pair  $(x_0, (w_t, \varepsilon_t)_{t \geq 0})$  and, by hypothesis,  $s_M$  is independent of that pair jointly,  $s_M$  is independent of  $Z_W$ . Joint independence is needed: independence of  $x_0$  and of the noise separately does not suffice (for instance, with  $x_0$  and  $w_0$  independent centred non-degenerate Gaussians,  $s_M = \text{sign}(x_0^\top w_0) v$  for a fixed  $v \neq 0$  is independent of  $x_0$  and of the noise separately, but not of the pair). Because  $s_M$  is drawn once per episode and persistent, the same vector  $s_M$  enters every  $t \geq \tau$  below.

*Output level.* Under Definition D2 the state recursion does not see the injection, so  $x_t = x_t^0$  and  $y_t = y_t^0 + \mathbb{I}[t \geq \tau] C s_M$ . Hence  $Y_W = Z_W + h_W(C s_M)$  with the linear map  $h_W(u) = (\mathbb{I}[t \geq \tau] u)_{t \in W}$ .

*State level.* Under Definition D9, before the onset the recursion does not see the injection, so  $x_\tau = x_\tau^0$ . For  $t \geq \tau$ ,  $x_{t+1} = A(x_t + s_M) + w_t$ ; with  $z_t = x_t - x_t^0$  this gives  $z_\tau = 0$  and  $z_{t+1} = Az_t + As_M$ , so  $z_{\tau+k} = \sum_{j=1}^k A^j s_M$  and  $\tilde{x}_{\tau+k} - x_{\tau+k}^0 = z_{\tau+k} + s_M = \sum_{j=0}^k A^j s_M$ . Hence on  $W = \{\tau, \dots, \tau+H-1\}$ ,  $Y_W = Z_W + G_H s_M$ , with  $G_H$  as in Theorem 3.

*Step 1 (convolution with a Gaussian is injective).* Let  $Z \sim \mathcal{N}(\mu, V)$  in  $\mathbb{R}^N$ ,  $V \succeq 0$ , and let  $U, U'$  be random vectors independent of  $Z$ . The characteristic function of  $Z + U$  is  $\varphi_Z(\omega) \varphi_U(\omega)$  with  $\varphi_Z(\omega) = \exp(i\omega^\top \mu - \frac{1}{2}\omega^\top V \omega)$ , which never vanishes (also for singular  $V$ ). So  $Z + U$  and  $Z + U'$  have the same law iff  $\varphi_U = \varphi_{U'}$  everywhere, i.e. (uniqueness theorem for characteristic functions) iff  $U$  and  $U'$  have the same law.

*Step 2 (linear images).* If  $L$  is linear and  $U, U'$  have the same law, so do  $LU, LU'$  (push-forward). If  $L$  has a linear left inverse  $L^+$  ( $L^+ L = I$ ), the

converse holds: apply  $L^+$ .

(i). Let  $W$  be finite and contain some  $t_0 \geq \tau$ . By Step 1,  $P_K^W = P_F^W$  iff  $h_W(Cs_K)$  and  $h_W(Cs_F)$  have the same law. The map  $h_W$  has the left inverse that reads the block at  $t_0$ , so by Step 2 this holds iff  $Cs_K$  and  $Cs_F$  have the same law. For infinite  $W$ , the law of  $Y_W$  is determined by its finite-dimensional marginals (Definition D3): if  $Cs_K, Cs_F$  have the same law, every finite sub-window has equal marginals (by the finite case, or trivially for a sub-window before the onset, whose law is that of  $Z$ ), so  $P_K^W = P_F^W$ ; conversely  $P_K^W = P_F^W$  gives equal marginals on the finite sub-window  $\{t_0\}$ , and the finite case applies to it.

(ii). By Step 1,  $P_K^W = P_F^W$  iff  $G_H s_K$  and  $G_H s_F$  have the same law. Row block  $k$  of  $G_H$  is  $C \sum_{j=0}^k A^j = \sum_{j=0}^k C A^j$ , the sum of row blocks  $0, \dots, k$  of  $\mathcal{O}_H$ ; hence  $G_H = (T_H \otimes I_m) \mathcal{O}_H$  with  $T_H$  the  $H \times H$  lower-triangular matrix of ones, which is invertible. By Step 2 (both directions,  $L = T_H \otimes I_m$  invertible), the condition is that  $\mathcal{O}_H s_K$  and  $\mathcal{O}_H s_F$  have the same law. For  $H \geq n$ : by the Cayley–Hamilton theorem each  $C A^j$ ,  $j \geq n$ , is a linear combination of  $C, C A, \dots, C A^{n-1}$ , so  $\mathcal{O}_H = M_H \mathcal{O}_n$  for a fixed matrix  $M_H$  whose first  $n$  row blocks form the identity; thus  $\mathcal{O}_H s = M_H(\mathcal{O}_n s)$  and  $\mathcal{O}_n s$  is the first  $n$  blocks of  $\mathcal{O}_H s$ , and by Step 2 equality in law of  $\mathcal{O}_H s_K, \mathcal{O}_H s_F$  is equivalent to that of  $\mathcal{O}_n s_K, \mathcal{O}_n s_F$ . For  $W = \{t : t \geq \tau\}$ : every finite sub-window lies in some  $\{\tau, \dots, \tau + H - 1\}$  with  $H \geq n$ , whose law determines that sub-window's marginal; so equality in law of  $\mathcal{O}_n s_K, \mathcal{O}_n s_F$  gives equal finite-dimensional marginals and  $P_K^W = P_F^W$ ; conversely  $P_K^W = P_F^W$  gives equality on the window with  $H = n$ .

(iii). If  $(A, C)$  is observable,  $\mathcal{O}_n$  has rank  $n$  (Definition D10) and hence a left inverse; by Step 2,  $\mathcal{O}_n s_K$  and  $\mathcal{O}_n s_F$  have the same law iff  $s_K$  and  $s_F$  do, and (ii) concludes.

*Point masses.* For  $s_M = \theta_M b_M$  deterministic, two point masses coincide iff the points do: (i) gives  $\theta_K C b_K = \theta_F C b_F$  (Theorem 1), (ii) gives  $\mathcal{O}_H \Delta_0 = 0$ , i.e.  $C A^j \Delta_0 = 0$  for  $j < H$ , equivalently  $G_H \Delta_0 = 0$  (Theorem 3), and on  $W = \{t : t \geq \tau\}$  gives  $\mathcal{O}_n \Delta_0 = 0$ , i.e.  $\Delta_0$  in the unobservable subspace (Theorem 2); (iii) gives  $\Delta_0 = 0$  (Corollary 3).  $\square$

#### Appendix A.2. Proof of Corollary 4

*Proof.* (i) Let  $\mathcal{O}(A) = [C; C A; \dots; C A^{n-1}]$ . Then  $U_C$  is the common zero set of the  $n \times n$  minors of  $\mathcal{O}(A)$ , polynomials in the entries of  $A$  (Definition D10). Since  $C \neq 0$ , it has a nonzero row  $c^\top$ . Extend  $v_1 = c^\top$  to a basis  $v_1, \dots, v_n$  of  $\mathbb{R}^{1 \times n}$  and define  $A_0$  by  $v_k A_0 = v_{k+1}$  for  $k < n$  and  $v_n A_0 = 0$ .

Then  $c^\top A_0^k = v_{k+1}$  for  $k < n$ , so the minor  $D$  formed by the rows  $c^\top A^k$  of  $\mathcal{O}(A)$  has  $D(A_0) \neq 0$ . The zero set of a polynomial that is not identically zero is Lebesgue-null, and  $U_C \subseteq \{D = 0\}$ , so  $U_C$  is null. (ii) For  $A \notin U_C$  the pair  $(A, C)$  is observable, and Corollary 3 gives the equivalence for every pair of instances;  $U_C$  is defined from  $A$  and  $C$  alone. A pair that Theorem 1 confounds with  $\theta_K b_K \neq \theta_F b_F$  fails the right-hand side, so it is distinguishable.  $\square$

### Appendix A.3. Proof of Proposition 4

*Proof. Step 1 (the two laws on  $W_H$ ).* With  $Z = Y_{W_H}^0$  the injection-free observation on  $W_H$  (Step 0 of the proof of Theorem 4 with  $s_M = \theta_M b_M$ ),  $Y_{W_H} = Z + G_H \theta_M b_M$  under state-level injection and  $Y_{W_H} = Z + \mathbf{1}_H \otimes C \theta_M b_M$  under output-level injection (every  $t \in W_H$  is post-onset).  $Z \sim \mathcal{N}(\mu^0, V_H)$  does not depend on the cause or on the model, so  $V_H$  is common. By (A1) and (A2),  $\varepsilon$  is independent of  $x^0$ , so  $V_H = \text{Cov}((Cx_t^0)_{t \in W_H}) + I_H \otimes \Sigma \succ 0$  by (A6). Hence  $Y_{W_H} \mid M \sim \mathcal{N}(\mu_M, V_H)$  with  $\mu_K - \mu_F = \delta_H$ , where  $\delta_H = G_H \Delta_0$  (state level) or  $\delta_H = \mathbf{1}_H \otimes C \Delta_0$  (output level), and  $d_H = \|V_H^{-1/2} \delta_H\|$  is the discriminability of Definition D5.

*Step 2 (closed form).* If  $d_H > 0$ , Proposition 3 applies to  $Y = Y_{W_H}$  and gives  $\text{CoA}_H^* = \text{CoA}_Y^*(d_H)$ . If  $d_H = 0$ , then  $\delta_H = 0$  ( $V_H \succ 0$ ) and the two laws coincide, so a policy  $\pi$  contains with the same probability  $\bar{\pi}$  under both causes,  $R(\pi) - R_{\text{oracle}} = a(1 - \bar{\pi}) + b\bar{\pi}$  (Definition D7), and the infimum over policies is  $\text{CoA}_H^* = \min(a, b)$ .

*Step 3 (a strict inequality).* Let  $g(d) = \text{CoA}_Y^*(d)$ , the closed form of Proposition 3, which tends to  $\min(a, b)$  as  $d \rightarrow 0^+$ . With  $u_\pm = \pm \log(b/a)/d - d/2$  and  $\varphi = \Phi'$ ,  $u_+^2 - u_-^2 = -2 \log(b/a)$ , so  $a\varphi(u_+) = b\varphi(u_-)$ ; differentiating,  $g'(d) = -a\varphi(u_+) < 0$ , so  $g$  is strictly decreasing on  $(0, \infty)$ . Hence  $g(d) < \min(a, b)$  for every  $d > 0$ . With Step 2:  $\text{CoA}_H^* < \min(a, b)$  iff  $d_H > 0$  iff  $\delta_H \neq 0$ .

(i).  $W_H \subset W_{H+1}$ , so every policy measurable in  $Y_{W_H}$  is a policy on  $W_{H+1}$  with the same expected cost (Definition D7); the infimum over the larger class is no larger,  $R_{H+1}^* \leq R_H^*$ , while  $R_{\text{oracle}}$  does not depend on  $H$ . Hence  $\text{CoA}_{H+1}^* \leq \text{CoA}_H^*$ . (Equivalently,  $d_H$  is non-decreasing:  $\delta_H$  and  $V_H$  are the leading sub-vector and block of  $\delta_{H+1}$  and  $V_{H+1}$ , and with  $S$  the Schur complement of  $V_H$  in  $V_{H+1}$ ,  $S \succ 0$ ,  $\delta_{H+1}^\top V_{H+1}^{-1} \delta_{H+1} = \delta_H^\top V_H^{-1} \delta_H + r^\top S^{-1} r \geq \delta_H^\top V_H^{-1} \delta_H$  for the appropriate residual  $r$ .)

(ii). Under output-level injection  $\delta_H = \mathbf{1}_H \otimes C \Delta_0$  is zero iff  $C \Delta_0 = 0$ , for every  $H$ . Step 2 and Step 3 conclude.

(iii). Under state-level injection  $\delta_H = G_H \Delta_0$ , which is zero iff  $CA^j \Delta_0 = 0$  for  $j = 0, \dots, H - 1$  (Theorem 3). Step 3 gives the equivalence. If  $(A, C)$  is observable and  $\Delta_0 \neq 0$ , then  $\Delta_0 \notin \bigcap_{k=0}^{n-1} \ker(CA^k) = \{0\}$ , so  $j^* = \min\{j : CA^j \Delta_0 \neq 0\}$  exists and  $j^* \leq n - 1$ ; for every  $H \geq h^* = j^* + 1 \leq n$ ,  $j^* < H$  and the condition holds.

*Final sentence.* If  $C\Delta_0 = 0$ , (ii) gives  $\min(a, b)$  at every horizon under output-level injection; under state-level injection  $j^* \geq 1$  because  $CA^0 \Delta_0 = C\Delta_0 = 0$ , so  $h^* \in \{2, \dots, n\}$ , and (iii) gives  $\text{CoA}_H^* < \min(a, b)$  for every  $H \geq h^*$ .  $\square$

## References

- [1] R. E. Kalman, On the general theory of control systems, IFAC Proceedings Volumes 1 (1) (1960) 491–502. doi:10.1016/S1474-6670(17)70094-8.
- [2] T. Kailath, Linear Systems, Prentice-Hall, Englewood Cliffs, NJ, 1980.
- [3] F. Pasqualetti, F. Dörfler, F. Bullo, Attack detection and identification in cyber-physical systems, IEEE Transactions on Automatic Control 58 (11) (2013) 2715–2729. doi:10.1109/TAC.2013.2266831.
- [4] E. Miehling, M. Rasouli, D. Teneketzis, A POMDP approach to the dynamic defense of large-scale cyber networks, IEEE Transactions on Information Forensics and Security 13 (10) (2018) 2490–2505. doi:10.1109/TIFS.2018.2819967.
- [5] S. Zonouz, H. Khurana, W. H. Sanders, T. M. Yardley, RRE: A game-theoretic intrusion response and recovery engine, IEEE Transactions on Parallel and Distributed Systems 25 (2) (2014) 395–406. doi:10.1109/TPDS.2013.211.
- [6] K. Hammar, R. Stadler, Intrusion prevention through optimal stopping, IEEE Transactions on Network and Service Management 19 (3) (2022) 2333–2348. doi:10.1109/TNSM.2022.3176781.
- [7] K. Hammar, R. Stadler, Learning near-optimal intrusion responses against dynamic attackers, IEEE Transactions on Network and Service Management 21 (1) (2024) 1158–1177. doi:10.1109/TNSM.2023.3293413.

- [8] K. Hammar, T. Li, R. Stadler, Q. Zhu, Adaptive security response strategies through conjectural online learning, *IEEE Transactions on Information Forensics and Security* 20 (2025) 4055–4070. doi:10.1109/TIFS.2025.3558600.
- [9] L. B. Booker, S. A. Musman, A model-based, decision-theoretic perspective on automated cyber response, *AAAI-20 Workshop on Artificial Intelligence for Cyber Security (AICS)* (2020). arXiv:2002.08957.
- [10] M.-A. Massoumnia, G. C. Verghese, A. S. Willsky, Failure detection and identification, *IEEE Transactions on Automatic Control* 34 (3) (1989) 316–321. doi:10.1109/9.16422.
- [11] D. Eriksson, E. Frisk, M. Krysander, A method for quantitative fault diagnosability analysis of stochastic linear descriptor models, *Automatica* 49 (6) (2013) 1591–1600. doi:10.1016/j.automatica.2013.02.045.
- [12] C.-Z. Bai, F. Pasqualetti, V. Gupta, Data-injection attacks in stochastic control systems: Detectability and performance tradeoffs, *Automatica* 82 (2017) 251–260. doi:10.1016/j.automatica.2017.04.047.
- [13] T. Roy, S. Dey, On distinguishability of anomalies as physical faults or actuation cyberattacks, *ASME Letters in Dynamic Systems and Control* 4 (3) (2024) 031003. doi:10.1115/1.4065292.
- [14] M. Taheri, K. Khorasani, I. Shames, N. Meskin, Cyberattack and machine-induced fault detection and isolation methodologies for cyber-physical systems, *IEEE Transactions on Control Systems Technology* 32 (2) (2024) 502–517. doi:10.1109/TCST.2023.3324870.
- [15] S. Zhou, C. Liu, J. Lan, X. Ren, X. Wang, Optimal distinguishable detection of faults and attacks in interconnected systems: A Stackelberg game approach, *Automatica* 193 (2026) 113226. doi:10.1016/j.automatica.2026.113226.
- [16] L. Zhou, L. Li, S. X. Ding, C. Louen, Observer-based fault-tolerant and resilient control under physical faults and integrity cyberattacks, *IEEE Transactions on Cybernetics* 56 (2) (2026) 763–773. doi:10.1109/TCYB.2025.3618684.

- [17] X. Xue, D. Shen, S. X. Ding, D. Zhao, Dual detection framework for faults and integrity attacks in cyber-physical control systems, arXiv preprint (2025). [arXiv:2510.14052](#).
- [18] L. Blackmore, B. Williams, Finite horizon control design for optimal model discrimination, in: Proceedings of the 44th IEEE Conference on Decision and Control and the European Control Conference (CDC-ECC), 2005, pp. 3795–3802. [doi:10.1109/CDC.2005.1582753](#).
- [19] J. Noom, O. Soloviev, C. Smith, M. Verhaegen, Online input design for discrimination of linear models using concave minimization, arXiv preprint (2024). [arXiv:2401.05782](#).
- [20] C.-W. Kong, J. McMahan, M. Lahijanian, Bayesian diagnosability and active fault identification, arXiv preprint (2025). [arXiv:2509.04708](#).
- [21] A. Anwar, A. N. Mahmood, Z. Shah, A data-driven approach to distinguish cyber-attacks from physical faults in a smart grid, in: Proceedings of the 24th ACM International Conference on Information and Knowledge Management (CIKM), 2015, pp. 1811–1814. [doi:10.1145/2806416.2806648](#).
- [22] N. Müller, K. Bao, J. Matthes, K. Heussen, CyPhERS: A cyber-physical event reasoning system providing real-time situational awareness for attack and fault response, *Computers in Industry* 151 (2023) 103982. [doi:10.1016/j.compind.2023.103982](#).
- [23] F. Ma, Y. Dang, Y. Liu, T. Zhou, Identifiability-aware, cost-aware triage of physical faults and measurement-integrity anomalies in energy cyber-physical systems, *Sensors* 26 (17) (2026) 5585. [doi:10.3390/s26175585](#).
- [24] A. A. Albustami, A. F. Taha, E. Bou-Harb, Unmasking stealthy attacks on nonlinear DAE models of power grids, *International Journal of Electrical Power & Energy Systems* 167 (2025) 110569. [doi:10.1016/j.ijepes.2025.110569](#).
- [25] R. Alisic, S. Amin, How sensor attacks transfer across Lie groups, preprint (2026). [arXiv:2604.03461](#).

- [26] P. Chullamonthon, P. Tangamchit, Ensemble of supervised and unsupervised deep neural networks for stock price manipulation detection, *Expert Systems with Applications* 220 (2023) 119698. doi:10.1016/j.eswa.2023.119698.
- [27] C. Yagemann, S. P. Chung, E. Uzun, S. Ragam, B. Saltaformaggio, W. Lee, On the feasibility of automating stock market manipulation, in: *Proceedings of the 36th Annual Computer Security Applications Conference (ACSAC)*, 2020, pp. 277–290. doi:10.1145/3427228.3427241.
- [28] S. J. Grossman, J. E. Stiglitz, On the impossibility of informationally efficient markets, *The American Economic Review* 70 (3) (1980) 393–408.
- [29] R. Xu, T. Zariphopoulou, L. Zhang, Decision making under costly sequential information acquisition: The paradigm of reversible and irreversible decisions, *arXiv preprint*, v1 posted 2023; v4 cited, 2026 (2026). arXiv:2401.00569v4.
- [30] L. Devroye, L. Györfi, G. Lugosi, *A Probabilistic Theory of Pattern Recognition*, Vol. 31 of *Stochastic Modelling and Applied Probability*, Springer, New York, 1996. doi:10.1007/978-1-4612-0711-5.
- [31] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I*, Wiley, New York, 1968.
- [32] D. Lopez-Paz, M. Oquab, Revisiting classifier two-sample tests, in: *International Conference on Learning Representations (ICLR)*, 2017. URL <https://arxiv.org/abs/1610.06545>
- [33] D. M. Green, J. A. Swets, *Signal Detection Theory and Psychophysics*, Wiley, New York, 1966.
- [34] H. V. Poor, *An Introduction to Signal Detection and Estimation*, 2nd Edition, Springer, 1994.
- [35] R. Cont, A. Kukanov, S. Stoikov, The price impact of order book events, *Journal of Financial Econometrics* 12 (1) (2014) 47–88. doi:10.1093/jjfinec/nbt003.

- [36] A. Teixeira, I. Shames, H. Sandberg, K. H. Johansson, A secure control framework for resource-limited adversaries, *Automatica* 51 (2015) 135–148. doi:10.1016/j.automatica.2014.10.067.
- [37] C. Elkan, The foundations of cost-sensitive learning, in: *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI)*, 2001, pp. 973–978.
- [38] D. Byrd, M. Hybinette, T. H. Balch, ABIDES: Towards high-fidelity multi-agent market simulation, in: *Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*, 2020, pp. 11–22. doi:10.1145/3384441.3395986.
- [39] R. Almgren, N. Chriss, Optimal execution of portfolio transactions, *Journal of Risk* 3 (2) (2001) 5–39. doi:10.21314/JOR.2001.041.
- [40] C. Mejrissi, Verification as a costed action: When checking beats containing under attack-fault ambiguity, working paper (2026).